

Subject Description Form

Subject Code	EE521
Subject Title	Industrial Power Electronics
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To provide power electronics engineers with in-depth knowledge of the industrial power electronics. 2. To provide latest development in power supplies, industrial power electronics system and their applications in renewable energy systems. 3. To give industrial concern in power electronics design including passive components and standards 4. To introduce to students to the various topologies of the power electronics circuits. 5. To enable students to understand the power quality issues and the active and reactive power flow.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Acquire a good understanding of power supply concept and design and be able to analyse the industrial needs for static power conversion. b. Understand the international standards on power electronics design. c. Have a global view on recent development on power electronics and be aware of applications of power electronics in various industries d. Understand the various topologies and working principles of basic power converters e. Work in teams and independently when conducting power electronics design and testing.
Subject Synopsis/ Indicative Syllabus	1. Industrial power systems: Static power systems, battery systems, AC systems, DC systems, AC-DC power conversion and recent advance in renewable energy systems such as wind and solar power 2. Power conversion: Soft-switching, power factor correction, inverter configurations and static converters. 3. Special environment power electronics: Power electronics distribution system, industrial guidelines, variable speed and constant frequency systems, actuation systems, brushless drives and other applications of power electronics in industry 4. Industrial power supplies: Converter topologies, decentralized power, power modules, electro-magnetic compatibility, international standards and reliability. 5. Power quality improvement: Fourier analysis of voltage current waveforms, total harmonic distortion, rectifier, passive/active filters, power quality issues, reactive power compensation. 6. Devices and Packaging: Hermetic and plastic packages, wire bonding, power devices, high temperature effect and substrates. 7. Magnetics and capacitors: High frequency inductors and transformers, winding techniques, core loss analysis, optimization of magnetics and power capacitors. Laboratory Experiments : Select at least one experiment from topics in DC-DC converter, power factor correction, power electronics for DC brushless motors, etc.

Teaching/Learning Methodology	Lectures and tutorials are the primary means of conveying the basic concepts and theories. Experiences on design and practical applications are given through experiments and mini-projects, in which the students are expected to solve design problems with real-life constraints and to attain pragmatic solutions with critical and analytical thinking. Interactive laboratory sessions are introduced to encourage better preparation and hence understanding of the experiments. Experiments are designed to supplement the lecturing materials so that the students are encouraged to take extra readings and to look for relevant information.																																																						
	Teaching/Learning Methodology		Outcomes																																																				
		a	b	c	d	e																																																	
	Lectures	✓	✓	✓	✓																																																		
	Tutorials	✓	✓	✓	✓																																																		
	Experiments/Laboratory	✓				✓																																																	
	Mini-project		✓	✓		✓																																																	
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><td rowspan="2">Specific assessment methods/tasks</td><td rowspan="2">% weighting</td><td colspan="5">Intended subject learning outcomes to be assessed</td></tr><tr><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td></tr><tr><td>1. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>2. Test and/or Assignment</td><td>20%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td></tr><tr><td>3. Laboratory performance & report</td><td>10%</td><td>✓</td><td></td><td></td><td>✓</td><td>✓</td></tr><tr><td>4. Mini-project & report</td><td>10%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100%</td><td colspan="5"></td></tr></table> One end-of-semester written examination; laboratory performance evaluation (including punctuality, initiative, and technical reasoning); laboratory report on a particular experiment; and mini-project of on a specific topic.							Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					a	b	c	d	e	1. Examination	60%	✓	✓	✓	✓		2. Test and/or Assignment	20%	✓	✓	✓	✓		3. Laboratory performance & report	10%	✓			✓	✓	4. Mini-project & report	10%	✓	✓	✓	✓	✓	Total	100%						
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Reading List and References	Reference books: 1. A. M. Trzynadlowski, Introduction to Modern Power Electronics, Third Edition, Wiley, 2015. 2. M.Cirrincione, M. Pucci, G. Vitale, Power Converters and AC Electrical Drives with Linear Neural Networks, CRC Press, 2012. 3. N. Mohan, Power Electronics: Converters, Applications, and Design, John Wiley & Sons, 2012. 4. G. M. Masters, Renewable and efficient electric power systems, John Wiley & Sons, 2004 5. K.W.E. Cheng, Classical Switched Mode and Resonant Power Converters, The Hong Kong Polytechnic University, 2002 6. Robert W. Erickson, Fundamentals of Power Electronics, Springer, 3rd edition, 2020																																																						